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AHA SCIENTIFIC STATEMENT

Single-Pill Combination Therapy for the Management of Hypertension: A Scientific Statement From the American Heart Association

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ABSTRACT: The growing global burden of hypertension and inadequate blood pressure control necessitate effective therapeutic strategies to improve blood pressure management and reduce the risks of cardiovascular diseases attributable to hypertension. Single-pill combination medications for hypertension combine ≥2 antihypertensive agents in a single tablet. Single-pill combination medications offer a promising opportunity to achieve faster and more sustained blood pressure control compared with stepped care (ie, prescribing antihypertensive monotherapy, titrating the dose, and later adding more antihypertensive agents). Single-pill combination medications combine complementary mechanisms of action of antihypertensive agents to more effectively lower blood pressure while reducing adverse effects. This approach simplifies treatment regimens by lowering pill burden, improves patient adherence, overcomes clinician inertia by simplifying prescribing, is cost-effective, and results in faster blood pressure control compared with using separate pills. Minor limitations of single-pill combination medications include constrained flexibility for dosing adjustments and lack of commercial availability of certain combinations of agents. By incorporating single-pill combination medications into routine practice for most patients with hypertension, health care professionals may improve long-term cardiovascular outcomes while reducing patient burden. This scientific statement provides an overview of the clinical evidence surrounding the use of single-pill combination medications for hypertension, strategies to implement single-pill combination medications into clinical practice, and knowledge gaps that merit further investigation.

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Key Words: AHA Scientific Statements ■ antihypertensive agents ■ cardiovascular diseases ■ cost-effectiveness analysis ■ drug combinations ■ hypertension ■ medication adherence

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Hypertension has long been recognized as a major modifiable risk factor for cardiovascular disease (CVD). Decades of epidemiologic studies and randomized clinical trials (RCTs) consistently demonstrate that lowering blood pressure (BP) reduces cardiovascular morbidity and mortality risk, including heart failure, myocardial infarction, and stroke.[1,2] The degree of cardiovascular risk reduction correlates closely with the magnitude of BP reduction.[3] Despite the availability of affordable and effective BP-lowering medications, hypertension remains a leading cause of morbidity and mortality globally.[4] In the United States alone, high BP accounts for hundreds of thousands of deaths per year and imposes a substantial economic burden on health systems and society.[5,6]

A core reason for this enduring public health crisis is that population-level BP control, defined as BP<130/80 mm Hg, remains stubbornly poor, at ≈25% in the United States.[7] One of the largest gaps in hypertension control exists among patients already receiving antihypertensive medications, where fewer than half of such individuals meet BP goals.[7] This reflects various factors, including patient-level barriers to medication use and adherence, clinician hesitancy to intensify treatment (ie, "therapeutic inertia"), and challenges such as patient costs due to insurance coverage or lack thereof, copayments, and limited formulary access to some medications.[1,2,8]

Single-pill combination medications (SPCMs), often referred to as fixed-dose combinations, simplify prescribing and patient adherence by packaging ≥2 active

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pharmaceutical ingredients in a single pill.[9–11] SPCM has gained attention as a potentially transformative strategy to improve longstanding gaps in BP control. In this scientific statement, we provide a review of SPCMs for hypertension and highlight practical strategies for incorporating SPCMs into clinical care. We also address gaps in knowledge that warrant future research.

COMBINATION ANTIHYPERTENSIVE THERAPY

Lifestyle interventions, such as diet modification, weight management, reduced sodium intake, and regular physical activity, are universally recommended as first-line measures for the prevention and treatment of hypertension. These nonpharmacologic approaches are foundational to BP control and cardiovascular risk reduction. However, the scope of this scientific statement is limited to pharmacologic management, with a particular focus on combination antihypertensive therapy.

Pharmacologic Principles and the Need for Combination Therapy

Combining antihypertensive medications generally produces fully additive effects on BP lowering when appropriate classes of agents are used together. However, the "law of the baseline value" (ie, the Wilder principle) needs to be considered. This physiologic principle underlies the empirical evidence showing that an equivalent dose of a medication will reduce systolic BP by 1 to 2 mm Hg less for every 10 mm Hg decrease in baseline systolic BP.[12–15] As such, combining 2 agents that each lower systolic BP by 10 mm Hg individually will produce a combined reduction of 18 to 19 mm Hg, rather than 20 mm Hg. Adding a second antihypertensive medication with a complementary mechanism of action results in a ≈5 times larger reduction in BP compared with doubling the dose of one drug, although exact values vary by population and drug class.[13] Because of the additive effects of agents addressing different physiologic mechanisms contributing to high BP, combination therapy often yields more uniform BP control across patient populations, irrespective of age, sex, race, or obesity.[16] Medication side effect rates also tend to be reduced by using lower-dose combinations versus higher-dose monotherapy.

Standard doses of single antihypertensive medications among all classes lower systolic BP by ≈10 mm Hg on average in a population.[13,17] There is, however, substantial variation in responses among medications within and between individuals. Given this limited efficacy and even accounting for variable responsiveness, estimates from multiple large-scale RCTs and longitudinal observational cohorts suggest that ≈75% of individuals with hypertension ultimately require ≥2 BP-lowering agents to maintain systolic/diastolic values below 140/90 mm Hg.[18] Achieving the more aggressive contemporary BP target of <130/80 mm Hg will therefore necessitate ≥2 medications for an even greater proportion of patients.[2] This was clearly illustrated in SPRINT (Systolic Blood Pressure Intervention Trial), which randomized participants to either standard systolic BP control (<140 mm Hg) or intensive control (<120 mm Hg). Participants randomized to the intensive BP control arm were prescribed an average of 2.8 medications, compared with 1.8 medications in the standard BP control arm, resulting in 80% of patients achieving a systolic BP <130 mm Hg.[19,20] Given the range of number of medications prescribed and final BP levels achieved, the trial findings suggest that most patients with hypertension will require between 2 and 4 medications.

SPRINT enrolled patients specifically at elevated cardiovascular risk. However, evidence from broader or lower-risk populations also supports that most patients with hypertension require multiple antihypertensive medications to achieve guideline-directed BP targets.[7] A clinical rule of thumb has been promoted since at least 2003 that 2-drug combinations should be started if BP is ≥20 mm Hg systolic or ≥10 mm Hg diastolic, or both, above goals. Several lines of evidence now suggest that this axiom should be reconsidered. First, more than half of all patients with systolic BP levels only 10 to 19 mm Hg above goal will require ≥2 medications to achieve goal BP. Second, intraindividual BP variability is substantial, and a single BP measurement "just below goal" could mean the individual is above goal some or most of the time (ie, average measured BP likely needs to be closer to 10 to 15 mm Hg below goal to be controlled most of the time). Third, achieving high population BP control <130/80 mm Hg rates (eg, 80% <130/80 mm Hg) also means that the "average" patient should have a mean BP closer to 10 to 15 mm Hg below that goal. Fourth, compelling data show that CVD risk reduction is monotonically proportional to the magnitude of systolic BP reduction achieved by treatment regardless of the starting BP level. The net health benefits of aggressive systolic BP lowering to levels below 120 to 125 mm Hg exceed the risks at the population level as well as for most individual patients. In this context, a treatment strategy focusing on prescribing the lowest possible dosage or number of medications with the explicit objective of achieving minimal BP control—defined as just below an arbitrary value (ie, <130/80 mm Hg)—does not accord with the latest evidence base and inadvertently serves to attenuate the full benefits of medical treatment. Fifth, given the variation in achieved BP reduction from antihypertensive treatment among patients, and within individuals among repeat visits, mean systolic BP levels in a population need to be at least 10 mm Hg lower than the goal threshold to achieve an 80% control rate.[21] These findings suggest that for most patients, even those with

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stage 1 hypertension (130–139/80–89 mmHg), 2 medications will likely be required to ensure that 80% of a population achieves BP goals below 130/80 mmHg.

A fundamental principle in combination therapy for hypertension management is that each medication should target a different physiologic mechanism contributing to elevated BP (eg, renin angiotensin system [RAS], sympathetic nervous system, sodium-volume control, arteriole vasoconstriction).[18] Over the years, several treatment protocols have been proposed.[16,22,23] However, current clinical guidelines preferentially recommend combinations including a RAS inhibitor paired with either a calcium channel blocker (CCB) or diuretic as first-line therapy, rather than β-blockers. Combinations such as 2 RAS inhibitors or a RAS inhibitor with a β-blocker are generally less effective for BP lowering and can increase the risk of adverse effects. Recent evidence suggests that combining a CCB with a diuretic or using 2 different diuretic types together (eg, a thiazide with a potassium-sparing diuretic) can be effective options.[24,25] In patients with resistant hypertension—those whose BP remains uncontrolled on optimal doses of 3 antihypertensive agents, including a diuretic—the addition of a fourth agent, typically spironolactone, is strongly recommended. The evidence regarding differences in health outcomes between combination regimens is discussed later in Clinical Applications.

Starting With Combination Therapy Versus Stepped Care

Stepped-care therapy (ie, initiating monotherapy and then titrating doses or adding new medications as needed to achieve BP control) was the primary strategy endorsed by major hypertension guidelines for several decades, particularly during the 1970s through 1990s.[26] However, more recent guidelines increasingly recommend starting with combination therapy—whether as separate pills or a SPCM—because it is safe, reduces therapeutic inertia, and leads to faster time to BP control.[11,27–30] Real-world data also suggest that starting a 2-drug SPCM may yield superior CVD risk reduction compared with initiating monotherapy (Figure 1).[31–37] Nevertheless, ≈40% of US adults with uncontrolled BP continue to receive monotherapy, a pattern that contributes to persistently poor population-level BP control and reflects the longstanding stepped-care prescribing practices, despite compelling evidence of the benefits of early combination therapy.[38]

SINGLE-PILL COMBINATION MEDICATIONS

Clinical Rationale and Current Status

Because most patients with hypertension require multiple medications to adequately manage BP levels, the resulting pill burden can negatively influence patient adherence. Poor adherence is particularly challenging in hypertension management, given that the condition is largely asymptomatic and patients do not experience immediate feedback or discomfort when missing medication doses. As a consequence, adherence tends to drop over time, particularly when patients are prescribed multiple individual pills or medications with complex dosing schedules.[39] SPCMs, which refer to formulations that combine ≥2 antihypertensive agents into a single pill, directly address these barriers by simplifying daily medication routines, reducing cognitive load around medication management, and improving patient adherence. SPCMs may also help mitigate clinician-level barriers to medication intensification by streamlining prescribing decisions. However, SPCMs could also potentially complicate subsequent treatment intensification decisions, especially when changes to individual components of an existing SPCM regimen are needed.

Despite these advantages, the uptake of SPCM in clinical practice remains relatively low. Analyses of National Health and Nutrition Examination Survey data suggest that ≈40% of US adults are prescribed monotherapy for hypertension, with little improvement between 2005 and 2016.[38] In fact, the age-adjusted prevalence of SPCM use decreased from 35.8% in 2009 through 2012 to 27.8% in 2017 through 2020 (P=0.01).[40]

Benefits of SPCMs Versus Free-Equivalent Combinations

Many expert guidelines for hypertension management worldwide—including the 2025 American Heart Association/American College of Cardiology, 2023 European Society of Hypertension, and 2024 European Society of Cardiology clinical practice guidelines—recommend the use of SPCM for patients who are taking or likely to need >1 medication for BP control.[1,2,41] This recommendation is supported by evidence demonstrating benefits of SPCMs versus free-equivalent combinations across several domains, including patient adherence, BP reduction, and CVD outcomes. Figure 2 provides a summary of the overall benefits versus limitations of SPCMs in the management of hypertension.

Patient Adherence and Persistence

A 2021 systematic review and meta-analysis by Parati et al[42] reported that SPCMs improved adherence in 18 of 23 included studies and increased persistence with therapy in 14 of 16 studies, compared with equivalent combinations administered as separate pills. Similar findings were replicated by Paoli et al[43] in a more recent 2024

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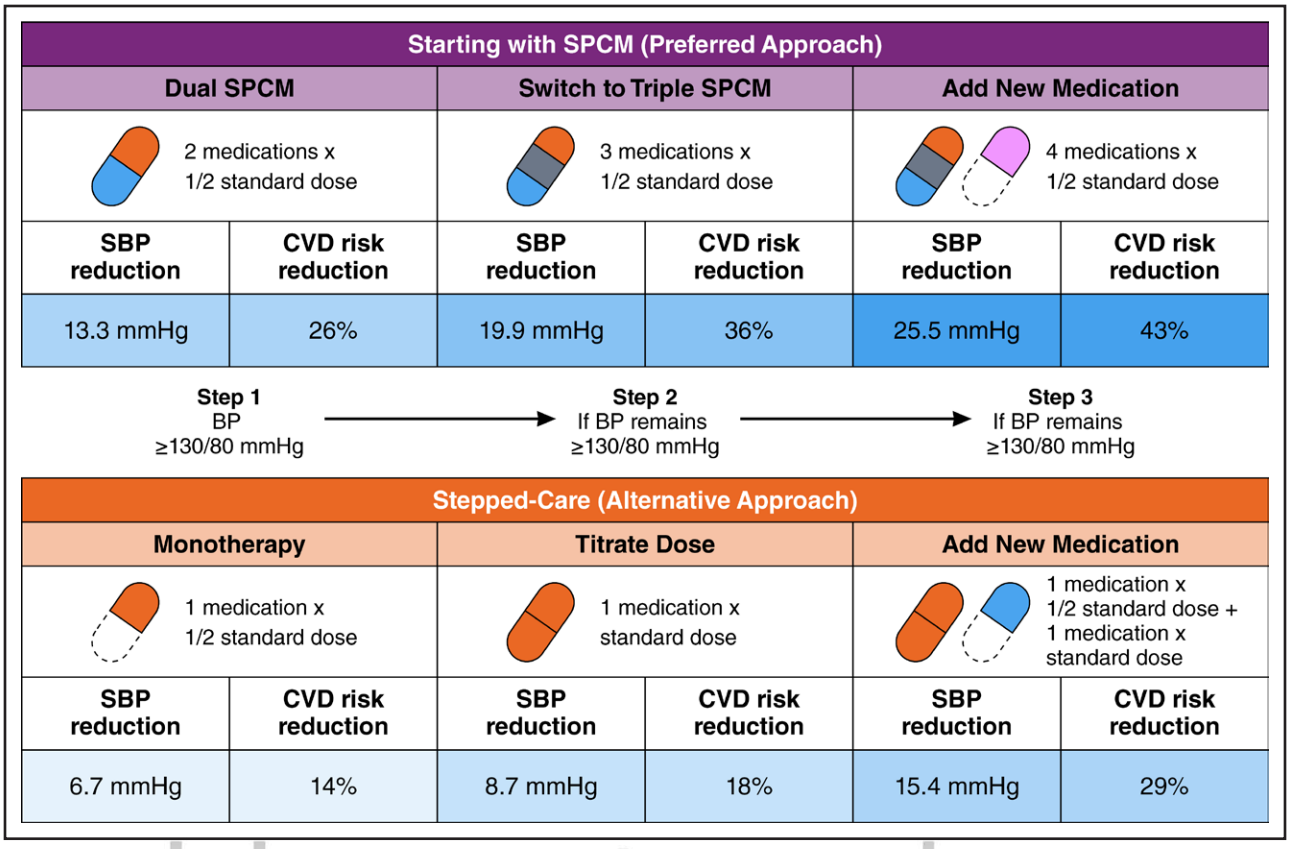

Figure 1. Starting with single-pill combination medications compared with stepped care.
The preferred approach of starting with single-pill combination medications (SPCMs) is more efficient, using fewer steps to achieve blood pressure (BP) control than standard stepped care. Another option for step 2 in the preferred approach (eg, BP only minimally above goal) would be to increase the dosages of 1 or both medications in the SPCM. Evidence supports using a renin-angiotensin system inhibitor combined with a calcium channel blocker as the initial combination in step 1 (see text for details). In step 3, evidence supports adding spironolactone as the 4th medication, even if a diuretic was introduced in step 2. The expected BP reductions presented in this figure are estimated for an individual with an untreated systolic BP of 150 mm Hg and are derived using the formulas provided in Law et al[31] and the meta-analytic findings from Ettehad et al.[32] The relationship between systolic BP reduction and major cardiovascular disease (CVD) risk is based on the relative risk of 0.80 per 10 mm Hg systolic BP reduction, as reported in the Ettehad et al[32] meta-analysis. These algorithms are simplified and indicative; additional or alternative steps may be required depending on patient-specific factors or different guideline recommendations. For example, current stepped-care guidelines recommending initial monotherapy often include additional titration steps before advancing to dual or triple therapy. SBP indicates systolic blood pressure.

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systematic review, reinforcing the adherence advantage of SPCMs.

BP Lowering and Time to Control

In addition to improved adherence, SPCMs lead to meaningful improvements in BP control. Parati et al[42] reported that SPCMs were also associated with modest but statistically significant reductions in both systolic BP (mean difference, −4.0 mm Hg [95% CI, −7.9 to −0.1]) and diastolic BP (mean difference, −1.5 mm Hg [95% CI, −2.7 to −0.4]) at 12 weeks compared with free-pill combinations. These findings were further supported by Paoli et al,[43] who similarly confirmed superior adherence and greater BP lowering favoring SPCMs. However, the generalizability of these results should be interpreted with caution; most studies included were small, short-term RCTs or observational studies involving patients with uncomplicated mild to moderate hypertension.

CVD Outcomes

Several observational studies have also associated SPCM use with a ≈15% to 30% lower incidence of major adverse cardiovascular events compared with equivalent free-pill combinations.[33–37] Reported benefits have been observed both when SPCMs were used to begin antihypertensive therapy as well as later during treatment when single or 2-drug regimens were changed or consolidated to SPCMs. The use of SPCMs for BP lowering has proven to provide outcome benefits in some clinical trials.[44] However, robust RCT data directly demonstrating that the usage of SPCMs per se improves cardiovascular outcomes versus a comparable regimen composed of individual pills in combination remain lacking.[45]

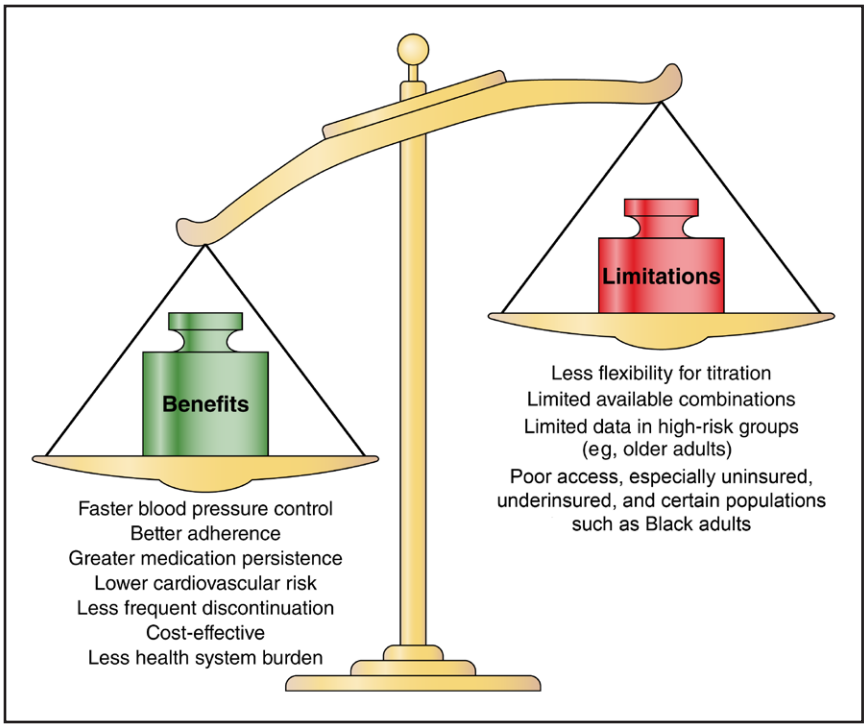

Figure 2. Benefits versus limitations of single-pill combination medications in the management of hypertension.
The overall evidence shows that the benefits of single-pill combination medications outweigh the limitations.

Cost-Effectiveness

Despite the potential for higher medication costs—particularly if the SPCM remains on patent or is placed on a higher patient cost-sharing tier—improvements in adherence, treatment intensification, and BP reduction associated with SPCM are projected in computer simulation studies to reduce CVD costs to the health sector (ie, all direct medical costs regardless of who pays them) by preventing CVD and CVD-related events.[41,46–54] Preventing these events is further estimated to increase quality of life, overall survival, and quality-adjusted life years, and to reduce CVD-related health care expenditures. As a result, recent studies characterize SPCM as either cost-effective (improved outcomes, including increased quality-adjusted life years, justify added costs) or cost-saving (both better outcomes and lower total costs).[41,48–54]

Nevertheless, variations in SPCM cost-effectiveness stem from differences in study design, including model type, clinical outcomes included, time horizon of the analysis, analytic perspective, specific costs considered, and populations studied. Some evaluations focus on the United States,[41,48] whereas others investigate settings in other countries.[49–54] In the United States, SPCM has been found to be either cost-saving or cost-effective for both initial therapy and when switching from separate pills.[41,48] One modeling analysis estimated that initiating SPCM with 2 antihypertensive agents in previously untreated adults could be cost-saving compared with stepped care and cost-effective—at an incremental cost of $60 600 (2024 USD) per quality-adjusted life years gained—when compared with free-equivalent combination therapy over 10 years.[41] Another study estimated that an even more aggressive early treatment strategy featuring an ultra-low-dose SPCM consisting of 4 antihypertensive medications was cost-effective compared with monotherapy in adults with hypertension in Australia.[49] In older US adults (age ≥65 years) with hypertension but no diabetes, switching to a 3-agent SPCM was estimated to be cost-effective at $39 700 (2024 USD) per quality-adjusted life years gained relative to free-equivalent combination therapy over a lifetime.[48]

These cost-effectiveness estimates are largely derived from short-term RCT data extrapolated through computer simulation models. Although these data are valuable for informing treatment decisions, the modeling approaches simplify clinical practice and rely on assumptions about treatment effects, medication adherence, and regimen adjustments. Alterations in modeling inputs, such as drug costs or projected adherence gains, may substantially shift SPCM cost-effectiveness estimates. For example, existing cost-effectiveness analyses have not incorporated potential long-term benefits of SPCMs on dementia prevention or cognitive decline, possibly underestimating the full societal and economic value of optimal BP management.

BARRIERS TO SPCM USE

Insurance Coverage and Reimbursement Policies

At the patient level, out-of-pocket cost is the most frequently cited patient barrier to SPCM use.[55] Insurance coverage, or lack thereof, greatly affects the use of

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SPCMs. Many SPCMs have been brand-name medications, resulting in higher costs and frequent exclusions from formularies, although most SPCMs are now available as generic products. However, generic availability does not automatically translate to patient affordability, especially when patient out-of-pocket costs remain substantial or when insurance coverage is limited or restrictive. Many insurers, including Medicare and Medicaid, continue to require step therapy or mandate the use of equivalent combinations prescribed as separate pills, despite the growing body of evidence demonstrating the cost-effectiveness of SPCMs.[40,46] Coverage for SPCMs varies across states, insurers, and managed care organizations, which exacerbates geographic disparities. Medicaid coverage is particularly restrictive, reflecting a tendency to favor separate generic pills.[56,57] Medicare Part D coverage varies among plans. An analysis by the Centers for Disease Control and Prevention showed that nonelderly adults on Medicaid had the lowest use of SPCMs and those on commercial insurance the highest.[58] From 2016 through 2020, SPCM use declined in both Medicare and Medicaid, although nearly all products were generic.[59] This practice disproportionately affects lower-income and underresourced populations, who are more likely to rely on Medicaid. In addition, financial incentives within the pharmaceutical supply chain, such as pharmacy dispensing fees associated with prescribing multiple individual pills, may contribute to lower SPCM use; however, this issue remains understudied and requires further investigation. The complex, often opaque, and sometimes irrational nature of drug pricing and reimbursement practices in the United States—including rebate structures, pharmacy benefit manager incentives, and formulary decisions—further complicates the adoption and use of SPCMs.[60]

Social Drivers of Health

There are substantial differences in the uptake of SPCMs across sociodemographic groups, highlighting disparities linked to socioeconomic status, insurance coverage, and geography. Lower-income and uninsured populations are less likely to be prescribed SPCMs, due to factors such as affordability barriers and inadequate access to health care professionals who are familiar with or willing to prescribe SPCMs.[61] Reviewing antihypertensive medications broadly, an analysis of Medicare Part D beneficiaries demonstrated higher rates of antihypertensive medication nonadherence among individuals with a low-income subsidy compared with those not receiving a subsidy (32.1% versus 25.4%).[62] The highest rates of antihypertensive nonadherence occurred in southern states. Coverage gaps disproportionately affect certain groups, such as Black patients, exacerbating existing racial disparities in hypertension control. Black adults face higher rates of uncontrolled hypertension compared with White adults, resulting in greater cardiovascular morbidity and mortality risk.[63] Inequities stem in large part from social drivers of health, which influence access to accurate diagnoses, timely treatment, and ongoing control of hypertension.[64–66] SPCMs represent a promising strategy to reduce such disparities, with randomized controlled trials suggesting potential benefits across all racial and ethnic groups.[67] However, access barriers must be addressed to achieve equitable SPCM adoption and improved hypertension outcomes.

Development, Manufacturing, Regulatory Approval, and Commercial Availability

SPCM options are modestly constrained by what is commercially available.[68] Among US adults prescribed ≥2 antihypertensive classes, ≈50% use class combinations not commercially available as SPCMs. Likewise, in the intensive arm of SPRINT, 60% of participants received a combination that lacks an SPCM equivalent.[68,69] These nuances could make it difficult to directly convert many existing 2- or 3-drug regimens into an exactly matching SPCM; however, the real-world relevance of this limitation is unclear. With the debatable exception of atenolol versus other β-blockers, no RCT evidence definitively shows CVD outcome benefits between specific generic agents within a medication class.[70] Regardless, β-blockers are not considered first-line therapy for hypertension due to inferior CVD risk reduction.[71,72] It is therefore also logical that starting an SPCM involving a β-blocker, even when combined with a first-line agent such as a thiazide diuretic, is a less favorable choice given that several SPCMs already exist that combine 2 or 3 first-line agents as recommended by guidelines (RAS inhibitors, CCBs, and thiazide diuretics).[73] Overall, there are ≈200 unique class combinations used by patients in the United States, but the 4 most commonly used combinations (RAS inhibitor/thiazide, RAS inhibitor/β-blocker, RAS inhibitor/CCB, and RAS inhibitor/thiazide/CCB) are available as SPCMs and represent approximately half of those receiving ≥2 medications.[69] In addition, real-world antihypertensive treatment patterns often deviate from guideline recommendations. Initiating or switching to an SPCM presents a valuable opportunity to improve these regimens by transitioning from medication classes or combinations lacking strong evidence or guideline endorsement to those with proven effectiveness. Therefore, the absence of an SPCM equivalent for a particular regimen frequently used in practice does not necessarily indicate a clinically meaningful gap. There is no commercially available SPCM combining a CCB with a thiazide-type diuretic, both of which are first-line classes.[2] The largest gaps in SPCM availability are in combinations of ≥3 medications and those with second-line agents, particularly spironolactone, an essential component in treating resistant hypertension. Expanding the range

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of available SPCMs could address some deficiencies, particularly triple and quadruple combinations, but the number of second-line agents limits the possibility of an SPCM for every situation.

Coformulating multiple pharmaceutical ingredients into a single pill introduces important challenges that are not always encountered with single–pharmaceutical ingredient medications.[74,75] These challenges start with selecting an optimal combination of medications. After determining that a regimen is clinically relevant, the individual components must also be verified to be chemically and physically compatible, ensuring consistent bioavailability of each ingredient in the presence of others. Additional hurdles can arise from the need to develop specialized manufacturing processes or formulation techniques, such as multilayer compression or coencapsulation methods, to accommodate distinct solubility or release profiles and prevent chemical degradation. In addition, certain CCBs commonly used in SPCMs (eg, nifedipine, verapamil, diltiazem) require extended-release formulations, adding complexity to formulation, manufacturing, and regulatory processes. Intellectual property issues, such as differing patent timelines for individual components of the SPCM, may further complicate the development process.

In the United States, the Food and Drug Administration (FDA) oversees the regulatory approval of SPCMs. In 2018, the FDA provided guidance to assist sponsors in the clinical development of SPCMs for the treatment of hypertension.[76] Each component of an SPCM must have a meaningful therapeutic effect. Usually, sponsors must conduct RCTs to demonstrate that the combination is superior in lowering BP compared with each single medication administered alone. The pharmacokinetics of each pharmaceutical ingredient in the combination must be characterized, and the FDA further recommends that SPCMs should be composed of previously approved antihypertensive drugs with distinct mechanisms of action. Typically, a single, large, phase III double-blind RCT is sufficient if it enrolls a population in which initiating therapy with multiple medications is appropriate and convincingly shows that the combination lowers both systolic and diastolic pressures more effectively than monotherapy. However, because the FDA requires evidence that each component of the SPCM contributes meaningfully to the therapeutic effect, demonstrating this within a single RCT would likely require a factorial design, potentially making it more challenging than conducting an additional separate RCT. Labeling for a newly approved product often includes figures illustrating the probability of achieving BP goals with the combination versus the individual products.

Regulatory requirements aim to ensure safety and efficacy, but they can also discourage the development of new SPCMs, especially if the constituent drugs are off patent, have minimal commercial profitability, or require intricate manufacturing. Although there is little evidence that specific medications within drug classes provide superior CVD risk reduction, some clinicians prefer specific agents (eg, chlorthalidone rather than hydrochlorothiazide) based on familiarity, a patient's comorbidity profile, or possible superior efficacy in certain populations at usual dosing ranges.[77] Although many SPCMs include commonly prescribed first-line drug classes, such as thiazide diuretics, RAS inhibitors, and CCBs, there may be situations (eg, side effects to first-line agents or comorbidities requiring indicated medications, like proteinuric chronic kidney disease) where the ideal therapeutic pairing is not offered in a single-pill formulation of 2 or even 3 medications. For example, fewer SPCMs in the United States incorporate β-blockers, thiazide-like diuretics (eg, chlorthalidone, indapamide), nondihydropyridine CCBs (eg, verapamil, diltiazem), or an aldosterone blocking agent (eg, spironolactone, eplerenone). Hence, currently available SPCMs lack the capacity to allow prescribing of every possible personalized regimen.[78] A major concern in the past was the limited availability of SPCMs incorporating chlorthalidone. Although chlorthalidone is generally more potent and longer-acting for BP lowering than hydrochlorothiazide at comparable doses, recent RCT results showing similar outcomes when BP is equally controlled limits this concern.[79]

Prescriber Knowledge, Attitudes, and Practices

Prescriber concerns about reduced flexibility in dose adjustments represent a frequent barrier to wider adoption of SPCMs. Some clinicians prefer separate pills because this allows easier adjustment or discontinuation of individual components if adverse events occur.[80] When using a combination regimen that consists of separate pills, a clinician has more options to uptitrate or downtitrate, or discontinue 1 medication at a time without potentially having to modify the rest of the regimen. However, this perceived barrier equating to the need to make refined medication or dosing changes (eg, to reduce side effects) must be weighed against the potential for negative consequences. Dosing changes common to real-world clinical practice (eg, from quarter to half standard dosage) only yield small alterations in BP (2–3 mm Hg) that are not clinically meaningful and are generally undetectable in any single patient (ie, below inherent BP variability and measurement accuracy).[13,14] Except in special circumstances (eg, frail patients), this practice has the potential to be counterproductive by increasing patient burden (eg, prescription refills, repeat visits) and delaying (or preventing) the achievement of BP goals. Similarly, initiating ≥2 medications simultaneously could make it more challenging to determine which drug is responsible for a side effect; however, this concern is not unique to SPCMs, and would apply to all combination therapy. This issue is particularly relevant among older adults, who may

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experience atypical presentations of medication side effects and are at increased risk for drug–drug interactions due to polypharmacy. Most drug classes have well-described causally related side effect profiles. Identifying the responsible component within a SPCM is therefore often straightforward (eg, edema for a CCB versus cough for an angiotensin-converting enzyme inhibitor). Whereas the seriousness of this concern may not be fully justified, it may nonetheless contribute to lower overall acceptance of SPCMs among clinicians. In addition, workflow issues, such as short appointment durations, limited awareness of available combination therapies, and low confidence in clinic-based BP measurements, have also been cited as barriers to SPCM use.[55]

CLINICAL APPLICATIONS

A large body of evidence reviewed in this scientific statement supports the overall favorable risk-to-benefit ratio of using SPCMs in the treatment of most patients with hypertension. The totality of evidence strongly supports starting 2-drug SPCM therapy rather than monotherapy for patients with stage 2 hypertension (ie, systolic BP ≥140 mm Hg or diastolic BP ≥90 mm Hg) and especially if systolic BP is ≥20 mm Hg above goal (ie, systolic BP ≥150 mm Hg). Our review of the evidence supports that 2-drug SPCM therapy has an overall favorable risk-to-benefit ratio when initiated as first-line therapy for all patients who require medication treatment for hypertension control regardless of the starting BP level, even in the range of 130–149/80–89 mm Hg. This includes low-risk patients with stage 1 hypertension not reaching BP goal after 3 to 6 months with lifestyle interventions alone and higher-risk patients in whom medications are indicated by the American Heart Association/American College of Cardiology high blood pressure guideline (eg, diabetes, chronic kidney disease, atherosclerotic cardiovascular disease). Only in a few select patients, such as those with only marginally elevated BPs (eg, 1–5 mm Hg above goal), and special scenarios (eg, frailty, orthostatic hypotension) should a more careful approach starting with monotherapy be considered. However, the extent to which many of these patients could equally benefit from and tolerate a low-dose SPCM remains to be better clarified.

The greatest clinical benefit of SPCMs is likely realized when they are used to appropriately intensify therapy (eg, transitioning patients from monotherapy to dual-drug SPCMs or from dual therapy to triple-drug SPCMs). If 2-drug SPCM treatment does not achieve BP control, the evidence supports changing to a triple SPCM (typically the addition of hydrochlorothiazide to baseline RAS inhibitor/CCB combination therapy) rather than adding a separate antihypertensive pill whenever possible. If BP remains uncontrolled on a 3-drug SPCM, workup to differentiate apparent from true resistant hypertension is warranted, similar to the approach used with individual-pill antihypertensive regimens.[81] In patients with true resistant hypertension, the evidence supports adding spironolactone as the most effective fourth antihypertensive agent.

Comparative-effectiveness data on specific combination regimens remain limited.[82] One of the few sources of available data is the ACCOMPLISH trial (Avoiding Cardiovascular Events Through Combination Therapy in Patients Living with Systolic Hypertension), the largest study to directly compare 2 combination regimens.[83] In the ACCOMPLISH trial, an SPCM of an angiotensin-converting enzyme inhibitor (benazepril) with a CCB (amlodipine) was 20% more effective at preventing the primary composite outcome of major CVD events compared with the combination of benazepril with a thiazide diuretic (hydrochlorothiazide). This benefit was independent of differences in BP reduction and was consistent across all subgroups.[82–84] Some have questioned whether using hydrochlorothiazide rather than chlorthalidone explains why the benazepril–amlodipine arm was superior in ACCOMPLISH.[85] However, the more recent Diuretic Comparison Project did not find substantial differences in major cardiovascular events favoring chlorthalidone, which largely negates this hypothesis.[79] The overall findings from ACCOMPLISH are further supported by positive trends observed in a second RCT, although likely underpowered, and by 2 separate meta-analyses.[86–88] Considering the totality of existing evidence, a compelling argument can be made that a RAS inhibitor plus a CCB should be strongly considered as the first-line 2-drug regimen for most patients with hypertension.[82]

Regimens that include second-line agents—such as β-blockers or loop diuretics—may merit reevaluation of the necessity of these therapies when BP is not adequately controlled or when simplifying therapy by converting to a SPCM. Alternative choices of using first-line agents in many of the available SPCMs should be strongly considered. In cases where a second-line agent remains necessary (eg, compelling indications such as a β-blocker for heart failure, or spironolactone for resistant hypertension) adding an SPCM to reduce the overall pill count and burden can still be beneficial. Clinical judgment is essential for scenarios involving intolerances to specific first-line agents or patients with complex comorbidities. In these instances, using separate agents may be unavoidable. Clinicians must remain attentive to insurance formularies and copayment structures that might limit patient access to SPCM. In situations where a patient's out-of-pocket costs are higher with an SPCM than with the individual agents, shared decision-making is essential. When feasible, low-cost or generic SPCMs should be advocated, particularly in resource-limited settings. A list of FDA-approved antihypertensive SPCMs is available in the 2025 American Heart Association/American College of Cardiology high blood pressure guideline.[2]

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From a health system perspective, accelerating the time to BP control through SPCM could reduce demands on clinical resources, such as repeat office visits or multiple titration steps.[89] A clear illustration of the real-world benefits of SPCM comes from the well-known Kaiser Permanente hypertension programs.[90–93] In the early 2000s, Kaiser Permanente Northern California and Kaiser Permanente Southern California each implemented multifaceted hypertension programs, which included support for SPCM use as part of simplified treatment algorithms. As SPCM utilization increased, so did the proportion of patients achieving guideline-directed BP control, increasing rates from <50% in the early 2000s to 90% (Kaiser Permanente Northern California) in 2013 and 85% (Kaiser Permanente Southern California) in 2012.

UNMET NEEDS AND FUTURE DIRECTIONS

There are several barriers and evidence gaps that need to be addressed to facilitate broader use of SPCMs (Table). For example, more triple and quadruple SPCM options are needed to address the considerable proportion of patients requiring ≥3 antihypertensive agents. Only 2 triple SPCMs are widely available in the United States, and no quadruple products have been approved, limiting the scope for further pill consolidation. In addition, many trials and meta-analyses have only included individuals with uncomplicated hypertension, underscoring the need for more comorbid populations in future clinical studies. Furthermore, implementation research is needed to develop strategies to improve clinician prescribing and patient uptake of SPCMs; the goal of implementation research is to understand how interventions work in different real-world settings. The uptake of newly developed SPCM formulations, even generic ones, may be limited by out-of-pocket costs and restrictive insurance formularies, potentially exacerbating existing socioeconomic and racial or ethnic disparities in hypertension outcomes.

Triple and Quadruple Medications

Recent and ongoing studies suggest that low-dose triple or quadruple SPCMs, such as 3 half-standard doses or 4 quarter-standard doses, can achieve more pronounced BP reductions than traditional stepped approaches, generally with acceptable side effects. A meta-analysis of 7 RCTs (n=1918) found that low-dose triple or quadruple SPCM reduced systolic BP by an additional 7.4 mm Hg (95% CI, 4.3–10.5) at 4 to 12 weeks compared with monotherapy or usual care, translating to a 20% higher rate of BP control (BP<140/90 mm Hg).[94] These improvements were sustained at 6 to 12 months, with low-dose triple or quadruple SPCM resulting in a further 6.4 mm Hg (95% CI, 1.8–11.0) reduction in systolic BP and a 13% higher rate of BP control. Although dizziness

Table. Knowledge Gaps and Future Directions for Research and Drug Development of Single-Pill Combination Medications for Hypertension

Address limited options of SPCMs
Develop more SPCM options that:
Include triple and quadruple drug combinations
Include CCB/thiazide diuretic drug combinations
Align with guideline-directed treatment and improve upon current real-world treatment patterns
Better address comorbidity-driven needs
Determine safety, tolerability, and effectiveness of consolidating ≥1 agents not provided in current formulations into a 2- or 3-drug SPCM containing alternative agents
Address limited evidence on existing SPCMs
Determine safety and efficacy of SPCMs in higher-risk populations, such as:
Resistant hypertension
Severe chronic kidney disease
Secondary hypertension (eg, primary aldosteronism)
Multiple comorbidities (eg, heart failure, diabetes)
Determine risk-to-benefit ratio of:
Initiating 2-drug SPCM in patients with stage 1 or stage 2 hypertension
SPCM in subsets of patients with physiologic susceptibility to adverse effects (eg, frailty, older age, preexisting orthostatic hypotension)
SPCM in populations experiencing structural barriers to care (eg, disenfranchised, underresourced populations)
Understand the utility of polypills
Research:
The pharmacologic feasibility of expanded polypill options
Dosing strategies for polypills
Develop polypills that:
Combine multiple antihypertensive agents with lipid-lowering therapy
Combine multiple antihypertensive agents with other CVD risk-reducing drugs
Address cost and formulary challenges, especially for uninsured or underinsured patients
Develop policy initiatives and payer collaborations aimed at:
Lowering copayments for SPCMs
Streamlining coverage processes for SPCMs
Broadening formulary inclusions for evidence-based SPCM options
Improve implementation of SPCMs into routine care
Identify whether there are additional clinician- or patient-level barriers to SPCM use
Improve clinician and patient education regarding the benefits of SPCMs
Use implementation science methods and frameworks to develop, test, and refine strategies to encourage appropriate step-up care, transitioning patients from monotherapy or dual therapy to dual- or triple-drug SPCMs
At the time of initiation of antihypertensive therapy
In patients on multiple separate antihypertensive agents
In patients who are uninsured or underinsured

CCB indicates calcium channel blocker; CVD, cardiovascular disease; and SPCM, single-pill combination medication.

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was slightly more common among those on SPCM (relative risk, 1.28 [95% CI, 1.00–1.63]), no association with serious adverse events or discontinuation was observed.

More recent RCTs produced similar findings. One RCT in Nigeria (n=300) reported that individuals started on a quarter-standard-dose triple SPCM experienced an additional reduction in mean home systolic BP of 5.8 mm Hg (95% CI, 3.6–8.0) and a 33% higher proportion achieving BP <130/80 mm Hg compared with standard care at 6 months.[95] Another RCT in the United States and other countries (n=295) found that both quarter-standard and half-standard triple SPCM reduced home systolic BP by 7.3 mm Hg (95% CI, 4.5–10.2) and 8.2 mm Hg (95% CI, 5.2–11.3), respectively, relative to placebo at 4 weeks, without a significant increase in serious adverse events.[96] In a retrospective analysis of a large cohort from Italy, triple SPCM versus 2 separate drugs (dual SPCM plus a third medication as a separate pill) was associated with superior medication adherence.[97] Taken together, these results suggest that initiating low-dose triple or quadruple SPCM and titrating doses upward could be more efficacious than a stepped-care approach for lowering systolic BP, and could possibly improve adherence and better prevent cardiovascular outcomes, without substantially increasing adverse event rates.

Despite these promising findings, the availability and use of triple or quadruple SPCMs in the United States remain limited. There are only 2 triple SPCMs on the market—olmesartan/amlodipine/hydrochlorothiazide and valsartan/amlodipine/hydrochlorothiazide—and no quadruple SPCMs are commercially available. Both triple SPCMs are available as generic formulations, potentially offering more affordable options. A new triple SPCM combining low doses of telmisartan, amlodipine, and indapamide is seeking regulatory approval by the FDA and may soon become available for first-line antihypertensive therapy in the United States.[96] In addition, a quadruple SPCM combining perindopril, amlodipine, indapamide, and bisoprolol is being developed.[98] Low-dose quadruple therapy could be practically implemented even with existing SPCMs. For example, combining half-tablets of commercially available SPCMs such as bisoprolol/hydrochlorothiazide and benazepril/amlodipine can effectively reduce the pill burden from 4 tablets to 2, potentially enhancing adherence while striving for the clinical efficacy associated with quadruple therapy.

Polypills

A related concept, the "polypill," extends SPCMs beyond multiple antihypertensive drugs by incorporating statins or aspirin into a single formulation.[99] This approach aims to comprehensively reduce cardiovascular risk for patients requiring multiple preventive therapies. An RCT conducted among 303 underresourced adults receiving care through a community health center in Mobile, Alabama, demonstrated the feasibility and effectiveness of this approach. The polypill (amlodipine 2.5 mg, losartan 25 mg, hydrochlorothiazide 12.5 mg, and atorvastatin 10 mg) led to greater 12-month reductions in systolic BP (net difference, −7 mm Hg [95% CI, −12 to −2]) and low-density lipoprotein cholesterol levels (net difference, −11 mg/dL [95% CI, −18 to −5]) compared with usual care, with high adherence (median 86%).[100] In a subsequent cost-effectiveness analysis, that polypill strategy was projected to be high-value care or result in cost savings.[101] These findings underscore the potential for polypill strategies to advance cardiovascular health and equity in underserved US populations. However, the same challenges that limit SPCMs—restricted availability and lack of suitable dosing options—hamper the adoption of cardiovascular polypills. To date, only 1 polypill for hypertension (amlodipine/atorvastatin) is commercially marketed in the United States, constraining how many patients can benefit from a simplified, all-in-one regimen. Ongoing efforts to develop new polypill formulations for the US market may ultimately help individuals manage their cardiovascular treatment while reducing daily pill burden.

CONCLUSIONS

Hypertension remains a pervasive public health problem despite the availability of effective, low-cost antihypertensive medications. SPCM provides a powerful tool to improve treatment adherence, accelerate BP control, and potentially reduce cardiovascular events. Evidence from RCTs and real-world data indicate that SPCMs can simplify regimens, can mitigate therapeutic inertia, and are cost-effective when modeled over the long term. The overall evidence favors the use of SPCMs as initial therapy (and as part of the treatment regimen) for most patients with hypertension. However, barriers to access—including affordability, insurance coverage, and clinician prescribing practices—need to be addressed, and greater awareness of the benefits among health care professionals is needed, to improve the widescale use of SPCMs in clinical practice moving forward.

ARTICLE INFORMATION

The American Heart Association makes every effort to avoid any actual or potential conflicts of interest that may arise as a result of an outside relationship or a personal, professional, or business interest of a member of the writing panel. Specifically, all members of the writing group are required to complete and submit a Disclosure Questionnaire showing all such relationships that might be perceived as real or potential conflicts of interest.

This advisory was approved by the American Heart Association Science Advisory and Coordinating Committee on September 22, 2025, and the American Heart Association Executive Committee on October 22, 2025. A copy of the document is available at https://professional.heart.org/statements by using either "Search for Guidelines & Statements" or the "Browse by Topic" area. To purchase additional reprints, call 215-356-2721 or email Meredith.Edelman@wolterskluwer.com

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Disclosures

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*Modest.

†Significant.

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